

Unit 1 - Math Fundamentals Series

MSDS 520

Meharry Medical College

Outline

Why Series?

A Problem to Motivate Us

A repeated dose

“A patient takes 100 mg of a drug every 6 hours. The drug’s half-life is also 6 hours, so half of what is in the body is gone by the time the next dose arrives. Does the drug pile up forever?”

Track the amount in the body just after each dose:

after dose 1: 100

after dose 2: $100 + 100(0.5)$

after dose 3: $100 + 100(0.5) + 100(0.5)^2$

after dose n : $100 + 100(0.5) + 100(0.5)^2 + \cdots + 100(0.5)^{n-1}$

Every dose adds another term, so the sum has more in it every day, and yet the answer is **no**, the total does not grow forever. Understanding *why* is what this lecture is about.

Where This Shows Up in Data Science

You are already summing things constantly

- **Every statistic is a sum:** a mean, a variance, a total count.
- **Every loss function is a sum:** add up how wrong the model was on each of the n training rows.
- **Every dot product is a sum:** the core operation of Unit 2, and of every neural network.
- **Computers only add:** e^x , $\sin x$ and $\ln x$ are all evaluated as sums of infinitely many simple terms, cut off partway.
- **Training is a sequence of steps:** “does this algorithm settle down?” is exactly the question of whether a series converges.

Unit 1-3 built **functions**. A series is what you get when you feed a function the numbers $1, 2, 3, \dots$ and add up the results.

What We Will Build

The plan

- ① **Sigma notation**: a compact way to write a long sum.
- ② **Closed forms**: finite sums that collapse to a single formula.
- ③ **Partial sums**: and what **convergence** means.
- ④ The **geometric series**, the one worth memorizing.
- ⑤ **Tests**: when does an infinite sum settle down, and when does it not?
- ⑥ Series as **approximations**, how a computer evaluates e^x .

Notebook for this lecture: U1-4_Series-1_Convergence.

Sigma Notation

Sequence First, Then Series

Sequence

A **sequence** is an ordered list of numbers, $a_1, a_2, a_3, \dots$, one number for each index. It is a function whose domain is the counting numbers: feed it n , it returns a_n .

Series

A **series** is what you get when you **add up** a sequence:

$$S = a_1 + a_2 + a_3 + \cdots + a_n$$

The distinction is worth keeping straight: a sequence is the *list*, a series is the *total*. In a data table, a column of values is the sequence; the number at the bottom of the column is the series.

Writing a Long Sum Compactly

Writing out $a_1 + a_2 + \cdots + a_{100}$ is impractical, and $a_1 + a_2 + \cdots$ forever is impossible. **Sigma notation** says the same thing in one symbol:

$$\sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \cdots + a_n$$

Reading the four parts

- $\sum$, the Greek capital sigma, meaning “add up what follows.”
- i , the **index**: a counter that takes one value at a time.
- $i = 1$ below and n above, where the counter **starts** and **stops**, both included.
- a_i , the **general term**: the recipe for the thing being added, written in terms of i .

Read it out loud as: “the sum, as i runs from 1 to n , of a_i .”

Sigma Is a For-Loop

The notation is exactly the loop you would write in code, start a counter, run to the limit, accumulate:

$$\sum_{i=1}^4 i^2$$

$$\begin{aligned} &= 1^2 + 2^2 + 3^2 + 4^2 \\ &= 1 + 4 + 9 + 16 = 30 \end{aligned}$$

```
total = 0
for i in range(1, 5):
    total += i**2
```

The upper limit is *inclusive* in mathematics and *exclusive* in Python's `range`, a mismatch that costs everyone a bug at least once.

The mechanical procedure

To expand a sigma: substitute the starting index into the general term, then the next value, and so on up to the upper limit, joining the results with plus signs.

The Index Is a Dummy Variable

The index has no meaning outside the sum, rename it and nothing changes:

$$\sum_{i=1}^n a_i = \sum_{j=1}^n a_j = \sum_{k=1}^n a_k$$

The starting value is a choice, not a law. These two write the same sum:

$$\underbrace{\sum_{i=1}^n 2^{i-1}}_{\text{start at 1}} = 1 + 2 + 4 + \cdots + 2^{n-1} = \underbrace{\sum_{i=0}^{n-1} 2^i}_{\text{start at 0}}$$

Shifting the start by one means shifting every appearance of i in the term to compensate. Starting at $i = 0$ is common for geometric series, because then the exponent i is the index.

Rules for Manipulating Sums

Three rules do almost all the work. Each is just regrouping an ordinary addition:

Constants pull out

$$\sum_{i=1}^n c a_i = c \sum_{i=1}^n a_i \quad \text{since} \quad c a_1 + c a_2 + \cdots = c(a_1 + a_2 + \cdots)$$

Sums split

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$$

A constant term counts itself

$$\sum_{i=1}^n c = \underbrace{c + c + \cdots + c}_{n \text{ times}} = nc$$

A Warning About What Does *Not* Split

The first two rules are called **linearity**, and linearity is about addition only. Products and powers do **not** pass through a sum:

$$\sum_{i=1}^n a_i b_i \neq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) \quad \sum_{i=1}^n a_i^2 \neq \left(\sum_{i=1}^n a_i \right)^2$$

Check it with the smallest possible example

Take $n = 2$ with $a = (1, 1)$ and $b = (1, 1)$. Then $\sum a_i b_i = 1 + 1 = 2$, while $(\sum a_i)(\sum b_i) = 2 \cdot 2 = 4$.

That gap is not a nuisance, it is the entire content of the variance formula in Unit 4, where $\sum x_i^2$ and $(\sum x_i)^2$ appear side by side and their difference measures the spread of the data.

Exercise: Reading Sigma Notation

Try it

- ① Write out every term of $\sum_{i=1}^3 \frac{a_i}{b_i}$.
- ② Write out every term of $\sum_{i=0}^5 \frac{i^2}{i+1}$, and find the total.
- ③ Rewrite $3 + 6 + 9 + 12 + 15$ in sigma notation.
- ④ Simplify $\sum_{i=1}^n (4a_i + 7)$ using the rules on the previous slides.

Exercise: Solution

1 and 2

$$\sum_{i=1}^3 \frac{a_i}{b_i} = \frac{a_1}{b_1} + \frac{a_2}{b_2} + \frac{a_3}{b_3}, \text{ the index rides along on both letters.}$$

$$\sum_{i=0}^5 \frac{i^2}{i+1} = \frac{0}{1} + \frac{1}{2} + \frac{4}{3} + \frac{9}{4} + \frac{16}{5} + \frac{25}{6} = \frac{687}{60} = 11.45, \text{ over a common denominator of 60. (Starting at } i=0 \text{ is legal, that term is simply zero.)}$$

3 and 4

Each term is 3 times its position and there are five of them: $\sum_{i=1}^5 3i$. (Equally correct: $\sum_{i=0}^4 (3i + 3)$.)

Split, pull the constant out, count the sevens: $\sum_{i=1}^n (4a_i + 7) = 4 \sum_{i=1}^n a_i + 7n$.

Sums You Already Use

Every one of these is sigma notation wearing a different name:

Four sums from the rest of this course

- **Mean** of n values: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ (Unit 4)
- **Sum of squared errors**: $E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ (Unit 3, optimization)
- **Dot product** of two lists: $\mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^p v_i w_i$ (Unit 2, vectors)
- **Linear model** with p features: $\hat{y} = b + \sum_{j=1}^p w_j x_j$ (everywhere)

Once you can read a sigma you can read nearly every definition in machine learning, the notation is how the field writes its ideas down.

The Other Big Operator: $\prod$

The same idea with multiplication instead of addition uses a capital pi:

$$\prod_{i=1}^n a_i = a_1 \cdot a_2 \cdot a_3 \cdots a_n \quad \text{for instance} \quad \prod_{i=1}^n i = n!$$

Logarithms turn products into sums

$$\ln \left(\prod_{i=1}^n a_i \right) = \sum_{i=1}^n \ln a_i$$

This is not a curiosity. Multiplying thousands of probabilities together collapses to zero on a computer (Unit 1-5), so statistics works with the **log-likelihood**, the $\sum \ln$ on the right, instead of the product on the left. The logarithm rules from Unit 1-3 are what make the swap legal.

Finite Sums with Closed Forms

Gauss's Trick

The story

Set the busywork of adding 1 through 100, a young Gauss wrote the answer down immediately. He had noticed that the list can be folded in half.

Write the sum forward, then backward underneath, and add column by column:

$$S = 1 + 2 + 3 + \cdots + n$$

$$S = n + (n - 1) + (n - 2) + \cdots + 1$$

$$2S = \underbrace{(n + 1) + (n + 1) + \cdots + (n + 1)}_{n \text{ columns, each equal to } n+1}$$

So $2S = n(n + 1)$, giving the **closed form**

$$\sum_{i=1}^n i = \frac{n(n + 1)}{2}$$

Why a Closed Form Is Worth Having

Two ways to get the same number

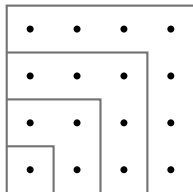
Loop: add n numbers one at a time, n operations. For $n = 10^9$ that is a coffee break.

Closed form: one multiplication and one division, the same work no matter how big n is.

A **closed form** is a formula that produces the total directly, with no loop and no dots. Finding one turns an n -step computation into a one-step computation.

The same instinct drives a great deal of numerical work: replace a long computation with a short formula that provably gives the same answer. When no exact closed form exists, we look for a good *approximation* instead, which is the second half of this lecture.

The Odd Numbers Make a Square



Each L-shaped shell adds the next odd number of dots, 1, then 3, then 5, then 7, and the finished picture is a 4×4 square.

So $1 + 3 + 5 + 7 = 4^2$, and in general

$$\sum_{i=1}^n (2i - 1) = n^2$$

Or derive it with the rules

$$\sum_{i=1}^n (2i - 1) = 2 \sum_{i=1}^n i - \sum_{i=1}^n 1 = 2 \cdot \frac{n(n+1)}{2} - n = n^2 + n - n = n^2$$

A Short Table of Closed Forms

Sum	Sigma form	Closed form
First n counting numbers	$\sum_{i=1}^n i$	$\frac{n(n+1)}{2}$
First n odd numbers	$\sum_{i=1}^n (2i-1)$	n^2
First n even numbers	$\sum_{i=1}^n 2i$	$n(n+1)$
First n squares	$\sum_{i=1}^n i^2$	$\frac{n(n+1)(2n+1)}{6}$

Only the first is worth memorizing, the others follow from it with the linearity rules, or can simply be looked up. What matters is knowing that such formulas *exist*.

The Finite Geometric Sum

A **geometric** sequence multiplies by the same **ratio** r at every step: $a, ar, ar^2, ar^3, \dots$. Its sum has a closed form too, found by a trick as slick as Gauss's.

Write the sum, multiply the whole thing by r , and subtract:

$$\begin{array}{rcl} S_n & = & a + ar + ar^2 + \dots + ar^{n-1} \\ rS_n & = & ar + ar^2 + \dots + ar^{n-1} + ar^n \\ \hline S_n - rS_n & = & a - ar^n \end{array}$$

Everything in the middle cancels. Factor and divide:

$$S_n(1 - r) = a(1 - r^n) \quad \implies \quad \boxed{\sum_{i=0}^{n-1} ar^i = a \frac{1 - r^n}{1 - r}} \quad (r \neq 1)$$

Exercise: Finite Sums

Try it

- ① Use the closed form to add $1 + 2 + 3 + \cdots + 100$.
- ② Add $2 + 4 + 6 + \cdots + 50$. (Careful: how many terms is that?)
- ③ A bacterial colony doubles every hour, starting from a single cell. Write the total number of cells present across the first 10 hours as a geometric sum, and evaluate it.
- ④ Why does the geometric formula exclude $r = 1$, and what is the sum in that case?

Exercise: Solution

1 and 2

$$\sum_{i=1}^{100} i = \frac{100 \cdot 101}{2} = 5050.$$

The terms are $2i$ for $i = 1, \dots, 25$, 25 of them, not 50: $\sum_{i=1}^{25} 2i = 25 \cdot 26 = 650$.

3. Geometric

With $a = 1$ and $r = 2$: $\sum_{i=0}^9 2^i = \frac{1 - 2^{10}}{1 - 2} = 1023$. Notice the shape, $2^{10} - 1$: doubling for n steps totals one less than the next power of two.

4. The excluded case

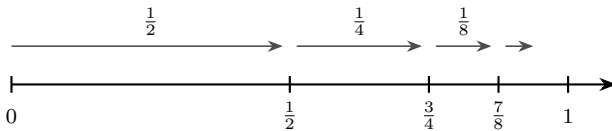
At $r = 1$ the derivation divides by $1 - r = 0$. But that case needs no formula: every term is just a , so n of them sum to na .

Infinite Series and Convergence

Zeno's Paradox

The puzzle, from about 450 BC

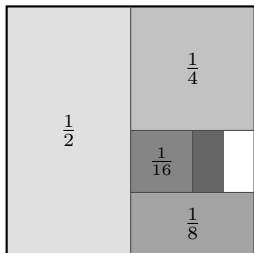
To cross a room you must first cover half the distance, then half of what remains, then half of *that* ... Since there are infinitely many steps, Zeno argued, you can never arrive.



Zeno was right that there are infinitely many steps. He was wrong that infinitely many positive numbers must add to infinity:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = 1$$

The Halving Series, in a Picture



The square has area 1. Shade half of it, then half of what is left, then half of *that*, forever.

No piece ever escapes the square, and no part of the square stays unshaded forever. The pieces are the terms; the square is the total.

So an infinite sum of positive numbers *can* be finite, provided the terms shrink **fast enough**. The rest of this section is about how fast is fast enough.

Partial Sums: How to Add Infinitely Many Things

We never actually add infinitely many numbers, we add the first n and watch what happens as n grows.

Partial sum

The n th **partial sum** of a series is the total of its first n terms,

$$S_n = \sum_{i=1}^n a_i$$

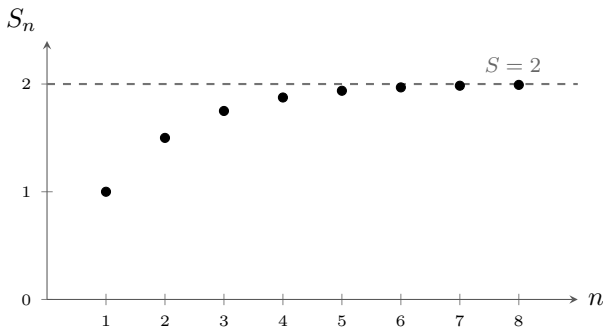
and the partial sums form a sequence of their own: $S_1, S_2, S_3, \dots$

Convergence

The series $\sum_{i=1}^{\infty} a_i$ **converges** to S if its partial sums settle on S : $S_n \rightarrow S$ as $n \rightarrow \infty$. If they settle on no number, it **diverges**.

An infinite sum is therefore not really an addition. It is a **limit** of ordinary finite additions.

Watching the Partial Sums Settle



Partial sums of $\sum_{i=0}^{\infty} \frac{1}{2^i} = 1 + \frac{1}{2} + \frac{1}{4} + \cdots$ Each term closes half of the remaining gap to 2, so the dots approach the dashed line without ever touching it. “The sum is 2” means exactly that: the gap can be made smaller than any tolerance you care to name, by taking enough terms.

The Infinite Geometric Series

Start from the finite formula and let n grow:

$$S_n = a \frac{1 - r^n}{1 - r}$$

Everything hinges on the single term r^n . When $|r| < 1$, repeated multiplication by r drives $r^n \rightarrow 0$, and the formula collapses:

$$\boxed{\sum_{i=0}^{\infty} ar^i = \frac{a}{1 - r}} \quad \text{provided } |r| < 1$$

Worth memorizing

This is the one infinite series whose exact value you should be able to write down on sight: **first term divided by one minus the ratio**.

Check it against the picture: $a = \frac{1}{2}$ and $r = \frac{1}{2}$ give $\frac{1/2}{1/2} = 1$, the whole square.

The Ratio Decides Everything

Ratio	Terms do what?	The series
$ r < 1$	shrink toward 0	converges to $\frac{a}{1-r}$
$r = 1$	stay the same size	diverges: $a + a + a + \dots$
$r = -1$	alternate, same size	diverges: the total flips forever
$ r > 1$	grow without bound	diverges

Sanity-check the formula's sign

Plugging $r = 2$ into $\frac{a}{1-r}$ returns $\frac{a}{-1} = -a$, a *negative* total for a sum of positive terms. That absurdity is the formula's way of telling you it was never valid there.

Always confirm $|r| < 1$ before using it.

Why $0.999\ldots = 1$

A repeating decimal is a geometric series in disguise. The digits of $0.999\ldots$ say

$$0.999\ldots = \frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + \cdots = \sum_{i=1}^{\infty} \frac{9}{10^i}$$

The first term is $a = \frac{9}{10}$ and the ratio is $r = \frac{1}{10}$, so $|r| < 1$ and the formula applies:

$$\sum_{i=1}^{\infty} \frac{9}{10^i} = \frac{9/10}{1 - 1/10} = \frac{9/10}{9/10} = 1$$

Not “almost” 1

These are two decimal spellings of one number, the way $\frac{1}{2}$ and $\frac{2}{4}$ are two fraction spellings of one number. Unit 1-2 showed that $\mathbb{R}$ has no gaps, if $0.999\ldots$ and 1 were different numbers, some third number would have to sit between them, and none does.

Exercise: Geometric Series

Try it

Decide whether each series converges, and if it does, find its exact sum.

① $\sum_{i=0}^{\infty} \frac{1}{2^i}$

② $\sum_{i=1}^{\infty} \frac{1}{3^i}$

③ $\sum_{i=0}^{\infty} \left(\frac{3}{2}\right)^i$

④ $\sum_{i=0}^{\infty} 5 \left(-\frac{1}{4}\right)^i$

For each one, identify a and r *first*, most mistakes here are bookkeeping mistakes about which term is the first one.

Exercise: Solution

1 and 2

(1) $a = 1$ (the $i = 0$ term) and $r = \frac{1}{2}$, so it converges to $\frac{1}{1 - 1/2} = 2$, the plot from a few slides back.

(2) This sum starts at $i = 1$, so the first term is $a = \frac{1}{3}$, not 1, with $r = \frac{1}{3}$:

$$\frac{\frac{1}{3}}{1 - 1/3} = \frac{1/3}{2/3} = \frac{1}{2}.$$

3 and 4

(3) Here $r = \frac{3}{2} > 1$, so the terms grow: it **diverges**, and no sum exists. (The formula would have returned -2 , nonsense, as promised.)

(4) $a = 5$ and $r = -\frac{1}{4}$, and $|r| = \frac{1}{4} < 1$, so it converges even though the terms alternate in sign: $\frac{5}{1 - (-1/4)} = \frac{5}{5/4} = 4$.

Terms Must Shrink: The n th-Term Test

The n th-term test for divergence

If a_n does **not** approach 0, then $\sum a_n$ **diverges**.

The reason is plain: if the terms keep some fixed size forever, then adding enough of them pushes the total past any bound you choose.

Read the direction carefully

This test can only ever prove **divergence**. Terms going to zero is **necessary** but not **sufficient**, exactly the logical distinction from Unit 1-1:

$$\text{converges} \implies a_n \rightarrow 0 \quad \text{but} \quad a_n \rightarrow 0 \not\implies \text{converges}$$

So $\sum \frac{n}{n+1}$ diverges, since its terms head for 1. But if $a_n \rightarrow 0$ the test says nothing at all, and the question stays open. The next slide shows that the gap is real.

Shrinking Is Not Enough: The Harmonic Series

The **harmonic series** has terms that plainly go to zero, and still diverges:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots = \infty$$

The grouping argument

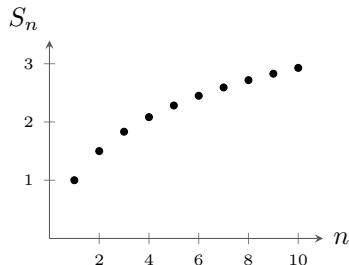
Bundle the terms in blocks of 1, 2, 4, 8, ... and replace every term in a block by the smallest one in it:

$$1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{> 2 \cdot \frac{1}{4} = \frac{1}{2}} + \underbrace{\left(\frac{1}{5} + \cdots + \frac{1}{8}\right)}_{> 4 \cdot \frac{1}{8} = \frac{1}{2}} + \underbrace{\left(\frac{1}{9} + \cdots + \frac{1}{16}\right)}_{> 8 \cdot \frac{1}{16} = \frac{1}{2}} + \cdots$$

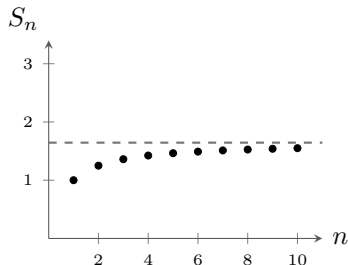
Every bundle contributes more than $\frac{1}{2}$, and there are infinitely many bundles, so the total exceeds any number you name.

It just does so *slowly*: getting past 10 takes more than 12,000 terms. Slow growth is still growth.

Two Series, Side by Side



$\sum \frac{1}{n}$, climbs forever



$\sum \frac{1}{n^2}$, settles at $\frac{\pi^2}{6}$

Both have terms heading to zero. Only the right-hand one shrinks fast enough for the total to stay finite, squaring the denominator is the whole difference between an infinite total and a finite one.

The p -Series Rule

The two examples above are the cases $p = 1$ and $p = 2$ of a single family:

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \quad \begin{cases} \text{converges} & \text{if } p > 1 \\ \text{diverges} & \text{if } p \leq 1 \end{cases}$$

The boundary is sharp

$\sum \frac{1}{n}$ diverges but $\sum \frac{1}{n^{1.01}}$ converges. Nothing dramatic happens to the individual terms at $p = 1$, the cutoff is about the **rate** of shrinking, not the size of any one term.

Two tests, in practice

- **Is it geometric?** Then $|r| < 1$ settles it, and you get the exact sum as a bonus.
- **Does it look like $1/n^p$?** Then $p > 1$ settles it, though it tells you nothing about the value.

Exercise: Converge or Diverge?

Try it

For each series, decide whether it converges, and say which test settles the question.

①
$$\sum_{n=1}^{\infty} \frac{n}{2n+1}$$

②
$$\sum_{n=1}^{\infty} \frac{1}{n^3}$$

③
$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

④
$$\sum_{n=1}^{\infty} (0.99)^n$$

Exercise: Solution

1 and 2

(1) As n grows, $\frac{n}{2n+1} \rightarrow \frac{1}{2} \neq 0$, so the n th-term test says it **diverges**. These terms never even get small.

(2) A p -series with $p = 3 > 1$: it **converges**. (The test says nothing about the value, and in fact no simple closed form for that value is known.)

3 and 4

(3) $\frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$ is a p -series with $p = \frac{1}{2} \leq 1$: it **diverges**, even though the terms do go to zero.

(4) Geometric with $r = 0.99$, and $|r| < 1$, so it **converges**. Starting at $n = 1$ makes the first term 0.99, so the sum is $\frac{0.99}{1-0.99} = 99$. A ratio just under 1 still converges, it simply takes an enormous number of terms to get there.

Series as Approximations

How Does a Computer Compute e^x ?

A processor can add, subtract, multiply and divide. It cannot “do” an exponential. So e^x is **defined** by an infinite series and **computed** as a partial sum:

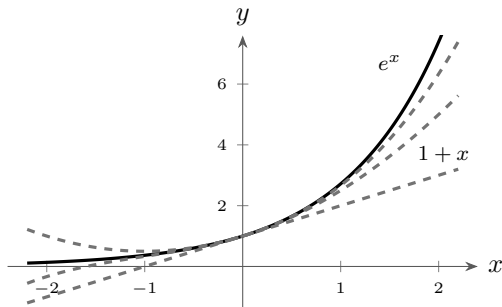
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$$

Why it converges so willingly

The factorial in the denominator eventually outruns any fixed x in the numerator, so the terms collapse toward zero very fast. That is why a handful of terms already gives full machine precision for modest x .

The same holds for $\sin$, $\cos$ and $\ln$. **Every transcendental function on your machine is a truncated series.** Where these particular coefficients come from is the subject of Taylor series, in Unit 3.

Truncations Closing In



The solid curve is e^x ; the dashed curves keep 2, 3 and 4 terms of the series. Each extra term hugs the true curve over a wider stretch. Near $x = 0$ even the straight line $1 + x$ is hard to distinguish from the curve, which is exactly the idea behind **linear approximation** in Unit 3.

Fast and Slow Convergence

Partial sums of $e = \sum_{n=0}^{\infty} \frac{1}{n!}$, whose true value is 2.718281828...

terms	1	2	3	4	6	9
S_n	1	2	2.5	$2.6\bar{6}$	$2.71\bar{6}$	2.71827877

Contrast Leibniz's beautiful but useless formula for π :

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots$$

Its error after n terms is roughly $\frac{1}{2n}$, so pinning down two decimal places takes hundreds of terms and ten decimals is hopeless. **Converging and converging usefully are different questions**, the first is mathematics, the second is engineering.

Where Do You Stop?

Truncation error

Cutting an infinite series off after n terms leaves behind everything you did not add:

$$\underbrace{S - S_n}_{\text{truncation error}} = \sum_{i=n+1}^{\infty} a_i$$

The practical stopping rule

Keep adding terms until the next one is smaller than the precision you need, or smaller than the precision the machine *has*. For a fast-converging series, the first omitted term is a good estimate of the error you are accepting.

There is a second, sneakier place where a sum stops. On a computer, adding a term that is tiny relative to the running total changes **nothing at all**, so the partial sums freeze even for a series that mathematically diverges. That is Unit 1-5, and it is why the harmonic series appears to converge in Python.

Exercise: Approximation

Try it

- ① Use three terms of $e^x = 1 + x + \frac{x^2}{2} + \dots$ to estimate $e^{0.1}$.
- ② The next omitted term is $\frac{x^3}{6}$. Roughly how large is the error you accepted?
- ③ Would those same three terms serve as well for e^5 ? Why or why not?
- ④ An algorithm removes half of its remaining error at every iteration, starting from an error of 1. Write the total error removed after n iterations as a series, and say what it converges to.

Exercise: Solution

1 and 2

$$1 + 0.1 + \frac{0.01}{2} = 1.105. \quad (\text{The true value is } 1.10517\dots)$$

The first omitted term is $\frac{0.1^3}{6} = \frac{0.001}{6} \approx 0.00017$, and the estimate is indeed off by about that much. For a rapidly shrinking series, **the first term you drop forecasts the error well.**

3 and 4

No. At $x = 5$ the first omitted term is $\frac{125}{6} \approx 21$, larger than the entire partial sum, so three terms are worthless there. The series still converges, but its terms grow before they shrink, and how many you need depends on x .

Total removed $= \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n} = \sum_{i=1}^n \frac{1}{2^i}$, which converges to $\frac{1/2}{1-1/2} = 1$: the error is driven to zero, but never reaches it in finitely many steps.

Back to the Dosing Problem

Answering the Opening Question

The setup, in this lecture's vocabulary

A dose of $D = 100$ mg every 6 hours, with a fraction $r = 0.5$ surviving until the next dose. Just after the n th dose the body holds

$$A_n = D + Dr + Dr^2 + \cdots + Dr^{n-1} = D \frac{1 - r^n}{1 - r}$$

, a **finite geometric sum**, exactly the one derived earlier.

Since $|r| < 1$, the term r^n dies away and the amount levels off at a **steady state**:

$$A_\infty = \frac{D}{1 - r} = \frac{100}{1 - 0.5} = 200 \text{ mg}$$

The drug does accumulate, but only to **twice** a single dose, never without bound. Each new dose is eventually offset by what the body clears, and the sum of infinitely many doses is finite.

What the Formula Tells a Clinician

$$A_{\infty} = \frac{D}{1-r}$$

- **Doubling the dose doubles the steady state:** D sits on top, linearly.
- **Shortening the interval is far more dangerous.** Dosing every 3 hours instead of 6 raises the surviving fraction to $r \approx 0.71$, and $\frac{100}{1-0.71} \approx 345$ mg, a jump of about 70% from a change that sounds modest.
- **As $r \rightarrow 1$ the steady state blows up.** A drug cleared slowly relative to its dosing interval accumulates toward toxicity; that is the mathematics behind “adjust the interval for renal impairment.”

One geometric series, three clinical readings. And the lesson generalizes: a repeated action with proportional decay always totals $\frac{\text{effect}}{1-\text{retention}}$, whether the effect is a drug dose, a dollar of spending, or a discounted future reward.

Review

Review

What we covered

- ① A **sequence** is a list; a **series** is its total. **Sigma notation** $\sum_{i=1}^n a_i$ is a for-loop written compactly.
- ② Sums are **linear**, constants pull out and sums split, but products and squares do *not* pass through.
- ③ **Closed forms** replace a loop with a formula: $\sum i = \frac{n(n+1)}{2}$, $\sum (2i - 1) = n^2$, and the geometric sum $a \frac{1-r^n}{1-r}$.
- ④ An infinite series is a **limit of partial sums**: it converges if S_n settles on a number, and diverges otherwise.
- ⑤ **Geometric series**: $\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}$ when $|r| < 1$, the one to memorize. It is also why $0.999 \dots = 1$.
- ⑥ Terms must go to 0 (the ***n*th-term test**), but that is not enough: the **harmonic series** diverges. For $\sum 1/n^p$, convergence needs $p > 1$.
- ⑦ Computers evaluate e^x , $\sin x$ and $\ln x$ as **truncated series**; where you stop is the **truncation error**.

Notebook: U1-4_Series-1_Convergence. **Previous:** Unit 1-3, Functions. **Next:** Unit 1-5, Floating-Point Arithmetic.